

# Recharge-Interception-root-zone model

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## 1 Intro

This recharge-interception root-zone model is meant to compute the recharge as percolation leaving the root zone under the condition that the root zone moisture content is above field capacity. In other words when the root-zone reservoir overflows.

The model first channels precipitation through its interception reservoir and then through its root-zone reservoir.

The equations that model the water budget and flow-through of these two reservoirs are derived next.

The interception reservoir is a bucket that receives all precipitation and from which water can evaporate freely as long as water remains in it. It's a bucket with a capacity  $S_{max}$  [mm], which is filled or emptied from some initial level and may become full or empty. It is emptied to the atmosphere when potential evaporation exceeds precipitation until it's empty. It's filled when precipitation exceeds potential evaporation and may fill up completely and overflow. The overflow enters the underlying root zone. Note that leaf area index is omitted. It can be added, but we will treat it as part of the size of the interception reservoir. If the leaf area index is less than 1, then the interception reservoir will just flow earlier into the root zone. So LAI does really not matter in the end as long as the size of the interception reservoir makes sense.

The root zone reservoir is similar, except that the evapo-transpiration, may be throttled when the actual filling of the root zone is below some limit. The throttling function is an important aspect of the root zone. The capacity of the root zone varies between the root-zone thickness times the lowest moisture value, i.e. the wilting point  $\theta_{wp}$  and the largest moisture value, i.e. field capacity  $\theta_{fc}$ . As soon as the root zone moisture content attains its  $\theta_{fc}$ , it can hold no additional water. The resulting overflow is then channeled downward into the percolation zone. Note that the evaporation from the root zone can never be larger than  $f(E_0 - E_I)$

where  $f$  is a crop factor,  $E_0$  is the open water evaporation and  $E_I$  the evaporation from the interception reservoir.

Modeling of the percolation zone itself is outside the scope of this document. It can be described by “Munsflow” or the “kinematic wave” approach or by modeling the unsaturated zone in full using an unsaturated zone model [Olsthoorn (2026)].

Ponding and surface runoff are ignored as considering a root zone as a model between precipitation, potential evaporation and the water table is only relevant for sandy areas with deep water tables where ponding may be regarded negligible on the longer run.

The time is divided into daily time steps that run from 08:00 to 08:00 the next day to match the time at which the daily precipitation values are recorded. We further split each day into two halves a) 08:00-20:00 and 20:00-08:00.

## 2 Interception reservoir

The water budget of the interception reservoir,  $S_I$ , is

$$\frac{dS_I}{dt} = P - E - q$$

There are no feed-backs. When simulating with a constant time step during which  $P$  and  $E$  are assumed constant and  $q = 0$  as long as the interception reservoir does not flow over. First consider  $P > E$ , so that the storage increases during the time step. At some time  $t = t_0 + \delta t$ , the storage will hit  $S_{max}$ :

$$S_{max} - S_0 = (P - E) \delta t$$

$$\delta t = \frac{S_{max} - S_0}{P - E}$$

When  $\delta t < \Delta t$  the the ceiling of the storage is reached within the time step  $\Delta t$  and will be kept to  $S = S_{max}$  afterwards. During the remainder of the time step the reservoir will overflow at a rate  $P - E$ . Therefore, the overflow  $q$  averaged over  $\Delta t$  will be

$$q = (P - E) \frac{\Delta t - \delta t}{\Delta t}; S = S_{max}$$

while  $P$  and  $E$  are not affected.

When  $\delta t > \Delta t$  then the reservoir will not be full at the end of the time step. We then set  $\delta t = \Delta t$  and the storage will then be

$$S = S_0 + (P - E) \Delta t$$

while  $q = 0$  with neither  $P$  nor  $E$  affected during the time step.

On the other head, when  $P < E$ , the reservoir empties. It would be empty at

$$\delta t = \frac{S_{min} - S_0}{P - E}$$

When  $\delta t < \Delta t$  the reservoir will be empty during the time step and we would have  $S = 0$  during the remainder. During this time,  $P$  remains unaffected while  $E$  is reduced to  $P$ . So the actual evaporation over the entire time step  $\Delta t$  can be composed of the values over the first and second part of the time step of, equally from the water budget over the entire time step

$$\begin{aligned} E_A &= E \frac{\delta t}{\Delta t} + P \frac{\Delta t - \delta t}{\Delta t} \\ &= P - \frac{S_{end} - S_0}{\Delta t}; S_{end} = 0 \\ &= P + \frac{S_0}{\Delta t} \end{aligned}$$

It is obvious that when the reservoir becomes empty during the time step  $E_A = P + \frac{S_0}{\Delta t}$

If  $\delta t > \Delta t$ , the reservoir will not become empty during the time step and, therefore, we can simply set  $\delta t = \Delta t$  and

$$S_{end} = S_0 + (P - E) \Delta t$$

during which the evaporation is unaffected. Essentially

$$E_A = E = P - \frac{S_{end} - S_0}{\Delta t}$$

The water budget over the entire time step will always be zero

$$B = P - \frac{S_{end} - S_0}{\Delta t} - E_A - q$$

During time steps in which  $P = E$ , nothing happens with the storage. And in case  $S_{max} = S_{min}$  there can be no storage at all, and the  $S$  remains the same and the interception reservoir is bypassed altogether. Setting  $S_{min} = S_{max}$  for the interception reservoir is, therefore, a way to bypass the interception without changing the overall recharge model.

### 3 Root-zone reservoir

To generalize the root-zone reservoir according to [Feddes (1982)], we divide it into two parts. The bottom part between  $0 \leq S \leq S_{crit}$  and the top part where  $S_{crit} < S \leq S_{max}$ . In the top reservoir there is no throttling of evaporation, while in the bottom part there is. The throttling in the bottom part of the root zone reservoir is linear. The top part of the root zone reservoir behaves exactly the same as the interception reservoir with the exception when it runs empty. If it runs empty during a time step, then the storage will become just lower than  $S_{crit}$  and will stay so for the remainder of the time step. We then use the linear throttling which depends of the fraction of the filling of the bottom part of the root zone reservoir.

#### 3.1 Bin model and earth model

The **bin model** has a root zone in which the evaporation is not throttled. It consists of the root zone top reservoir only. The **earth model** has a double root zone, split at storage levels  $S_{crit_R}$ . Below this level the throttling is linear. Below we just derive the equations for the unthrottled model and for the linearly throttled model. We'll just use the unthrottled model (root zone top reservoir) for the separate bin model. The bin model is equivalent to the earth model when we set  $S_{crit_R} = 0$  in the earth model. By setting  $S_{crit_R} = S_{max_R}$  we cancel the unthrottled top reservoir from the root zone, so that the entire root zone will be throttled linearly. The latter model and the bin model can be made equivalent to the lambda model derived further down, by setting  $\lambda = 1$  and  $\lambda = 0$  respectively.

#### 3.2 Root zone top reservoir

So we may start the time step with  $S > S_{crit}$  and therefore we are in the top reservoir. If  $P > E$  it will fill up exactly the same way as the interception reservoir did. And if the top reservoir is full during the time step at  $t_0 + \delta t$ , it will overflow during the remainder yielding a recharge  $q$  with a time-step averaged value equal to

$$q = (P - E) \frac{\Delta t - \delta t}{\Delta t}$$

Like with the interception reservoir if  $\delta t > \Delta t$  then the reservoir will not become full during the time step  $\Delta t$  so that  $q = 0$  and we set  $\delta t = \Delta t$  to compute the storage at the end of the time step

$$S_{end} = S_0 + (P - E) \Delta t$$

When however,  $P < E$  the top reservoir empties. It will be empty at

$$\delta t = \frac{S_{crit} - S_0}{P - E}$$

Note that because in this case both the denominator and the numerator are less than zero,  $\delta t > 0$ . If  $\delta t > \Delta t$ , the top reservoir will not become empty during the time step. The then set  $\delta t = \Delta t$  and compute  $S$  at the end of the time step

$$S = S_0 + (P - E) \Delta t$$

If, however,  $\delta t < \Delta t$ , the top reservoir does become empty during the time step. So we have

$$S_{crit} = S_0 + (P - E) \delta t$$

Then, for the remainder of the time step,  $\Delta t = \delta t$ , the storage will be smaller than  $S_{crit}$  and we are in the bottom reservoir and handle the situation accordingly.

### 3.3 Root zone bottom reservoir

We may start in the bottom reservoir when  $S < S_{crit}$  at the beginning of the time step or land in the bottom reservoir from the top reservoir when  $P < E$ .

In the bottom reservoir  $E$  is no longer constant it is reduced by the filling fraction:

$$E_a = E \frac{S}{S_{crit}}$$

The lower reservoir's budget is

$$\frac{dS}{dt} = P - E \frac{S}{S_{crit}}$$

We can ignore the overflow  $q$  for the bottom reservoir never overflows since when  $S$  passes  $S_{crit}$  when  $P > E$ , we transfer from the bottom to the top reservoir from the moment it happens during the rest of the time step. We can now solve the differential equation for the situation when  $P$  and  $E$  are both constant and  $0 \leq S \leq S_{crit}$ .

To integrate the ODE we apply the standard procedure.

$$-\frac{S_{max}}{E} \frac{d\left(P - E \frac{S}{S_{crit}}\right)}{dt} = P - E \frac{S}{S_{crit}}$$

with  $\zeta = P - \frac{E}{S_{max}} S$ , this simplifies to

$$\frac{d\zeta}{\zeta} = -\frac{E}{S_{crit}} dt$$

Integrating both sides gives

$$\ln \zeta = -\frac{E}{S_{crit}} t + C$$

The integration constant follows from  $\zeta = \zeta_0$  at  $t = t_0$ , yielding

$$\ln \frac{\zeta}{\zeta_0} = -\frac{E}{S_{crit}} (t - t_0)$$

Taking the exponent at both sides results in

$$\zeta = \zeta_0 \exp\left(-\frac{E}{S_{crit}} (t - t_0)\right)$$

Replacing  $y$  with its original expression then yields

$$P - E \frac{S}{S_{crit}} = \left( P - E \frac{S_0}{S_{crit}} \right) \exp \left( - \frac{t - t_0}{S_{crit}/E} \right)$$

After division by  $E$  we have dimensionless parameters. Setting  $y = \frac{S}{S_{crit}}$  and setting  $T = S_{crit}/E$ , finally gives the expression to work with

$$y = \frac{P}{E} + \left( y_0 - \frac{P}{E} \right) \exp \left( - \frac{t - t_0}{T} \right) \quad (1)$$

When  $E > 0$  and, we have  $T > 0$  is the characteristic time of this reservoir. The exponent will then always approach 0 for  $t \rightarrow \infty$ . Hence, the steady-state solution is

$$y = \frac{S}{S_{crit}} = \frac{P}{E} \rightarrow P = E \frac{S}{S_{crit}}$$

that is, when  $P$  equals the throttled evaporation.

Of course, when  $E = 0$  there can be no throttling of  $E$  and the bottom reservoir becomes linear. We have to capture such time steps, because the expression 1 will otherwise break down.

When  $P < E$  the bottom reservoir empties. But due to the throttling, it will never become completely empty. The situation at the end of the time step can then always be computed using expression 1 with  $t - t_0 = \Delta t$ .

When  $P > E$ , the bottom reservoir fills up according to expression 1. The time at which this would occur follows from

$$t = t_0 - T \ln \left( \frac{y - \frac{P}{E}}{y_0 - \frac{P}{E}} \right)$$

for  $x = \frac{S}{S_{crit}} = 1$ . Hence

$$\delta t_1 = T \ln \left( \frac{y_0 - \frac{P}{E}}{1 - \frac{P}{E}} \right) \text{ with } y_0 = \frac{S_0}{S_{crit}}$$

Note that, because for  $P > E$  and  $x_0 < 1$  both the denominator and numerator of this expression are less than zero, so that  $\delta t > 0$ .

Like explained before, if  $\delta t_1 > \Delta t$  then the reservoir will not be full during the time step, we then set  $\delta t_1 = \Delta t$  and compute  $x$  at the end of the time step by setting  $t - t_0 = \Delta t$  in expression 1. However, if  $\delta t_1 < \Delta t$ , then the bottom reservoir will become full within the time step, and we have  $S = S_{crit}$  at  $t = t_0 + \delta t_1$ . At that moment we transfer the case to the bottom reservoir for the remainder of the time step  $\Delta t - \delta t$ , starting with  $S = S_{crit}$ .

We can cancel the bottom reservoir by setting  $S_{crit} = 0$ . If we then hit the bottom of the reservoir and land in the bottom reservoir with zero storage, we pass the case on with zero change for the remainder of the reservoir. We then ignore the remainder of the time step, and calculate the actual evaporation over the entire time step from its water budget

$$E_a = P - (S_{end} - S_0) / \Delta t$$

which is always correct. With the same logic we may cancel the top reservoir by setting  $S_{crit} = S_{max}$ . When we enter the top reservoir from the bottom reservoir in this case, the top reservoir will just overflow for the remainder of the time step, which is exactly what should happen.

## 4 Lambda model

### 4.1 Throttling the evaporation using $(S/S_{max})^\lambda$

A more general approach would be to throttle the evaporation like so

$$E_a = E \frac{S}{S_{max}}$$

in which  $E$  is constant and  $E_a$  is the actual evaporation. Then for  $\lambda = 0$  we have the model with unrestrained, constant evaporation and with  $\lambda = 1$  we have the model in which the evaporation is throttled linearly as derived above. But we now have more flexibility in the extent we throttle the evaporation. On the other hand we do away with the two-reservoir concept elaborated above. In practice we would choose a values for  $\lambda$  between 0 and 1, so that the model will behave between these two extremes and will always show more throttling with lower filling of the root zone reservoir.

The general case for  $0 \leq \lambda \leq 1$ , defined the rate of change of the storage  $S$  in the root zone by

$$\frac{dS}{dt} = P - \left( \frac{S}{S_{max}} \right)^\lambda E$$

so that with  $0 \leq x = S/S_{max} \leq 1$ ,  $p = P/S_{max}$  and  $r = E/S_{max}$ , all positive numbers, we have

$$\frac{dy}{dt} = p - y^\lambda r$$

So that the required integration

$$y(t) = p(t - t_0) - r \int_{t_0}^t y^\lambda dt$$

needs to be carried out to get a solution. Although it's possible to do that analytically, it becomes complicated and difficult to handle special cases. We therefore, prefer to integrate numerically, choosing the familiar Runge-Kutta method in the scipy library. This numerical method also allows recording the times at which given thresholds are met. The only threshold we care about is when with  $P > E$  the storage hits  $S_{max}$  after which the reservoir overflows and produces the sought recharge. We should be careful to prevent numerical overshooting and undershooting.

## 4.2 Scipy's initial values ODE solver `scipy.integrate.solve_ivp`

Scipy's integrate initial value solver '`scipy.integrate.solve_ivp`' solves a system of ODE's. It numerically integrates a system of ordinary differential equations given an initial value:

$$\begin{aligned} \frac{dy}{dt} &= f(t, y) \\ y(t_0) &= y_0 \end{aligned}$$

In which  $y$  may be a vector and  $t$  is the 1D independent parameters. The solver requires the **function**  $y = p - rx^\lambda$  which must be a callable, a **time span**  $t = (0, \Delta t)$ , a **method**, for which we choose 'RK45' and **events** to track along the way and save the times of occurrence, and **arguments** for the function. And  $y(t_0)$ , which is  $S/S_{max}$  at  $t = 0$ , a parameter **dense\_output**, which if True computes a continuous solution.

The events that are tracked are functions for which it's tracked when they cross zero. Each even can have a **terminal** argument, which if True makes the integration stop. Each event can also have a **direction** argument which when +1 tracks only when the event function changes from negative and when -1 when the event function crosses zero from negative to positive.

The event function to track overshooting  $y = 1$  is  $event_1 = y - 1$  with *direction* = +1 and the event function to track undershooting is  $event_2 = y$  met *direction* = -1.

## 5 Water budget

The interception model accepts precipitation  $P_0$ , potential evaporation  $E_0$ , and the current storage state of the interception reservoir  $S_{I0}$ . Its output is the state of the interception reservoir at the end of the time step  $S_I$ , the amount that reaches the ground  $P_1$  and the actual evaporation from the interception reservoir  $E_I$ . The water budget of the interception reservoir during each time step in the code reads:

$$B_I = P_0 - E_I - (S_I - S_{I0}) / \Delta t - P_1 = 0$$

The sum of the right-hand terms is zero, which is also verified during the simulation.

Then follows the root zone, which starts with its current status  $S_{R0}$ , and accepts  $P_1$  and the remainder of the potential evaporation  $E_0 - E_I$ . What comes out is the state of the root zone reservoir at the end of the time step  $S_R$  the recharge  $rch$  which is the overflow of the root zone reservoir averaged over the time step and  $E_r$  the actual evaporation from the root zone. The water budget for the root zone reads

$$B_R = P_1 - (E_0 - E_I) - (S_R - S_{R0}) / \Delta t - rch = 0$$

The sum of the right-hand terms also equals zero in every time step and is verified during the simulation. The overall water budget equals  $B_I + B_R = 0$ , so we have

$$\begin{aligned} B_I + B_R &= P_0 - E_I - (S_I - S_{I0}) / \Delta t - P_1 + P_1 - E_R + (S_R - S_{R0}) / \Delta t - rch \\ 0 &= P_0 - (E_I + E_R) - ((S_I - S_{I0}) + (S_R - S_{R0})) / \Delta t - rch \end{aligned}$$

So we have precipitation  $P_0$ , overall actual evaporation  $E_A = E_I + E_R$ , the total storage change  $(S_I - S_{I0} + S_R - S_{R0}) / \Delta t$  and recharge  $rch$

The model starts with a `pd.DataFrame` with fields “RH” and “EV24” which hold the precipitation and potential evaporation provided by the weather station converted to mm/d. The outcome is the same `DataFrame` but with more columns holding the results for each time step. Currently the time steps are generally 1D, but in fact, any time step size will work, because the length of the time step is obtained from the `datetime` index of the data frame. The added columns are

**IC** In volume that was interecepte during the time step:  $IC = (P_0 - P_I) \Delta t$  [mm]

**STO** The storage level at the end of the time step  $STO = S_I + S_R$  [mm]

**EA** The actual evaporation during the time step:  $EA = (E_I + E_R) \Delta t$  [mm]

**RCH** The recharge during the time step:  $RCH = rch \times \Delta t$  [mm]

**BI** The interception water budget:  $BI = (P_0 - E_I - (S_I - S_{I0}) / \Delta t - P_I) \Delta t$  [mm] (must be 0)

**BR** The root zone water budget:  $BR = (P_I - E_R - (S_R - S_{R0}) / \Delta t - rch) \Delta t$  [mm] (must be 0)

Therefore, all columns IC, STO, EA, RCH, BI and BR have dimension [mm] over the time step, while the storage is in mm valid for the end of the time step, and, therefore for the beginning of the next time step. This implies that the precipitation columns RH and the potential evaporation columns EV24 must also be in mm over the time step. This will be automatically true for daily values from weather stations and time steps of 1 d. However the model will also work with any other time step but then the user must assure that P and EV24 must be the accumulated values during the chosen time step. In fact, the model will even work should the time step length vary from time step to time step.

## 6 Example

Fig.1 shows the result of three root-zone recharge models with the daily precipitation and Makkink (well-watered grass) evaporation of De Bilt as input. The bin model has no throttling of the evaporation, the earth model has linear throttling when  $S < S_{crit_R}$  and the lambda model has throttling of the evaporation

according to  $E_a = E \left( \frac{S}{S_{max_R}} \right)^\lambda$ . The values for  $S_{max_I}$ ,  $Scrit_R$  and  $S_{max_R}$  and  $\lambda$  are given in the header of the figure. The top figure shows the original EV24 (Makkink data) and the computed actual evapotranspiration  $E_A$ , which, of course is the throttled EV24. The second axes show the precipitation and the actual recharge and the third axes shows the storage of the root zone over time for the three models. It's clear that the storage values and, therefore, the  $E_A$  and recharge values do not differ much between the models.

The models can be run with  $S_{max_I} = 0$ , which effectively cancels the interception reservoir. Furthermore, setting  $Scrit_R = 0$  cancels throttling of the evaporation in the **earth model** until the root zone is completely empty, making this version of the **earth model** behave exactly like the **bin model**. Lastly, the  $Scrit_R$  can be set equal to  $S_{max_R}$  which cancels the unrestrained top reservoir of the **earth model** making the throttling of the evaporation linear all along the entire root zone, which is equivalent to using  $\lambda = 1$  in the **lambda model**. Note that the **bin model** is equivalent to using  $\lambda = 0$  in the **lambda model**.

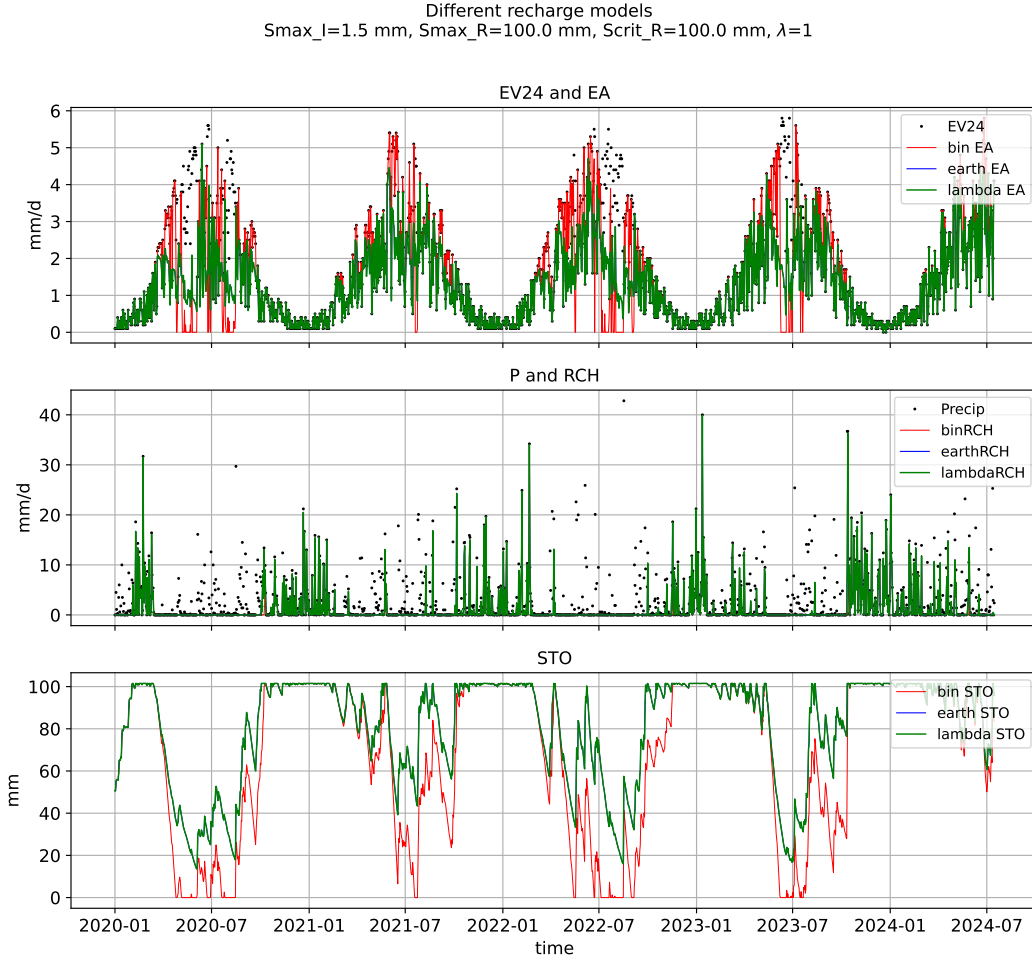
### 6.0.1 Throttling in the lambda model

Fig.2 shows the  $y = E_a/E = (S/S_{max_R})^\lambda$  as a function of the degree of filling of the root zone for different values of  $\lambda$ . For  $\lambda = 0$  (not shown) we always have  $E_a = E$ . For  $\lambda=1$  throttling is linear, so  $E_a/E$  increases linearly with the degree of filling. For  $0 < \lambda < 1$  the throttling increases progressively with emptying of the root zone. In practice, judging from the curves a values of  $\lambda$  between say 0.1 and 0.25 may well resemble the real-world relation. Which, of course, should be verified on the hand of real-world data. Linear throttling or throttling in accordance with  $\lambda = 0.5$  intuitively seems too mild as evaporation will be hardly reduced as long as the degree of filling is above 0.5 to 0.7. This is why in the earth model, we split the root zone into two parts, an upper part with a filling between  $Scrit_R$  and  $S_{max_R}$  in which no throttling takes place and a lower part with filling between 0 and  $Scrit_R$  in which the throttling is linear. This way the lambda model and the earth model can be made to react more or less the same for most practical situations. The earth model in which the  $Scrit_R$  plays a role is the model mostly using in practice.

## References

- [Feddes (1982)] Feddes, R. A. (1982). Simulation of field water use and crop yield. In F. W. T. Penning de Vries, & H. H. van Laar (Eds.), Simulation of plant growth and crop production (pp. 194-209). (Simulation monographs). Pudoc. <https://edepot.wur.nl/172222>.
- [Olsthoorn (2026)] Olsthoorn T.N. (2026) Percolatie door dikke onverzadigde zones - Munsflow versus Kinematic Wave. Stromingen 2026/01, p17-31





root\_zone\_rch\_example.ipynb | 2026-05-18 00:25:10

Figure 1: Example using precipitation and Makkink evaporation from De Bilt weather station together with the actual evaporation, recharge and running root-zone storage of the three models. The dots show the data. During summer when the root zone is not full there is not recharge and so the dots showing the actual precipitation “fly” in the second axes, while in the winter whenever the root zone overflows, and evaporation is almost zero, precipitation and recharge are almost the same. Similarly, whenever the root zone reservoir is low, actual evaporation is much lower than the potential evaporation shown by the dots in the upper axes. The lower axes shows that the root zone empties fully for the unthrottled bin model but neither for the earth model nor for the lambda model. Notice that the interception and root-zone data are shown in the heading of the figure.

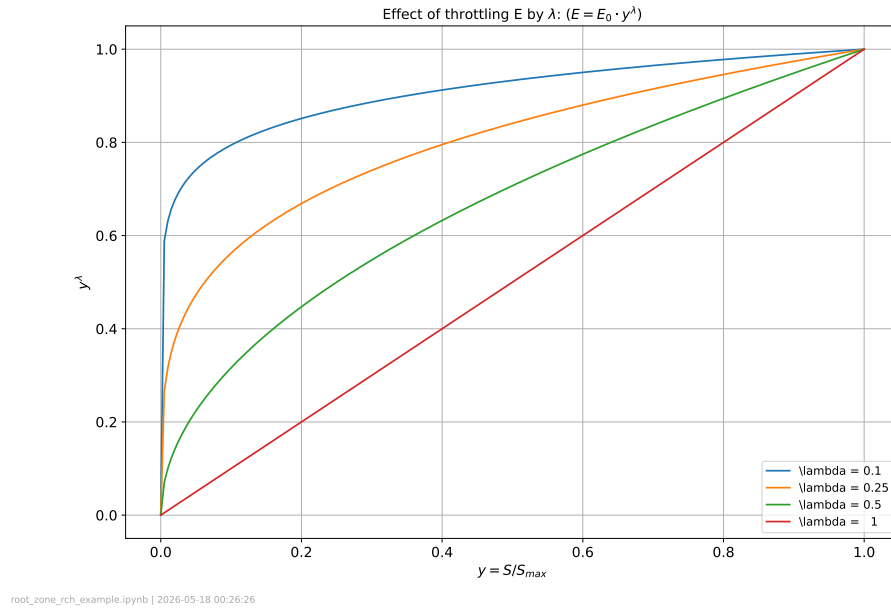


Figure 2: Remaining evaporation as fraction of potential evaporation as a function of the degree of filling of the root zone for different values of  $\lambda$ .